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Executive Summary

This report is a critical assessment of the use of cloud security mechanisms to help in solving data integrity and identity management issues in the e-commerce contexts. It presents the prominent risks to cloud security such as data breach, insider threats, and DDoS attacks, as well as insecure API, and suggests the multilayer approach that involves encryption, IAM, AI-driven monitoring, and Zero Trust systems. The analysis also suggests that effective governance, ethical data treatment, and ongoing surveillance measures can maximise organizational resilience and guarantee that the organisation is in compliance with GDPR and ISO 27001. In general, the report is practical and based on research studies to attain secure, reliable, and compliant cloud infrastructure. On the one hand, in the future, AI-based surveillance and quantum resistant encryption will transform the next wave of safe cloud systems to allow predictive defense and moral automation.

Introduction

Cloud computing has revolutionized the way organizations deal with IT infrastructure by offering a scalable storage system, flexible computing resource and economies of scale. Cloud platforms are useful to e-commerce companies because they can serve a large amount of data related to customers, process transactions, and provide services.

Nevertheless, when sensitive information, including customer data and financial data, is migrated to the cloud, it poses a great security risk. Lessen threats include information breaches, insider assaults, and Distributed Denial-of-Service (DDoS) assaults that are becoming more often. Organizations are also expected to adhere to such policies and laws as GDPR and PCI DSS, in which well-developed security is essential to ensure data integrity and customer trust (Karkuzhali et al., 2024).

In order to reduce these risks, a multi-layered security mechanism such as encryption, identity and access management, continuous monitoring and incident response plan should be adopted in the businesses. This report assesses cloud security alternatives to resolve organizational and technical issues, determines the most significant risks and threats, and suggests the strategies to ensure the safety of data in transit, at rest, and in use .

Task 1: Evaluate the Application of Cloud Security

Task 1.1 Challenges and Importance of Data Integrity

The significance and difficulties of data integrity are explained and justified by challenges.

Data integrity refers to the guarantee that data is correct, consistent and untouched through its lifecycle of existence, which spans between creation, storage and processing, and retrieval. In cloud computing, it is very important to maintain data integrity because cloud computing is distributed and dynamic. E-commerce firms, especially, depend on quality of information in processing of orders, stock records, and financial statements. Any data breach can cause the loss of finances, fines, and loss of customer confidence.

22 There are a number of threats to data integrity in cloud environments:

- **Data Replication and Distribution:** Data replication in cloud systems is a common occurrence whereby a single data centre can be replicated to various servers, data centres or even geographic locations to provide high availability and recovery in times of disaster. Although this enhances reliability, it brings about the risk of inconsistencies among copies of a file where one copy could be outdated or corrupted than the other. A sophisticated replication scheme and integrity checking system is needed to ensure that every copy is kept in sync and correct.
- **Unauthorized Access or Modification:** Cloud environments can be accessed via the internet, which exposes the available cloud to threats of malicious individuals. Without strict access control, sensitive data can be modified, deleted or corrupted by attackers or even malicious insiders. The changes made illegally might not be identified, unless the monitoring and auditing systems are very strong, which may jeopardise business activities and legal requirements.
- **Transmission Errors:** There is always transmission of data between clients and the applications on cloud servers. Data can be corrupted during transmission due to network interruptions, latency problems or software mistakes. In the absence of integrity verification tools like checksums or cryptographic hash functions, corrupted data might be stored or manipulated without any knowledge and influence the business decisions or customer transactions.

Ensuring the integrity of data is particularly important regarding the adherence to regulatory frameworks, including GDPR and PCI DSS that presuppose businesses to secure personal data and financial information. Lacking the assurance of data integrity might mean legal action, loss of money and reputation of the customers.

6 The issue of data integrity in e-commerce systems is not just a technical issue, but it is also a legal and ethical requirement. Customer confidence and compliance with rules determine the correctness and integrity of data. Consequently, encryption, hashing and access-control should be creatively incorporated to provide end-to-end assurance to data in transit, data at rest and data in use. All these measures enhance business continuity and adherence to the regulations of GDPR and PCI DSS. Although encryption does help in reducing technical risks like interception or

corruption, its success is reliant on organizational controls like key-management, employee competence and compliance audits proving the integrity is not only a technical concern, but a governance concern as well.

Task 1.2 Cloud Security Options to Ensure Data Integrity

The concept of data integrity in cloud environments is used to describe the correctness of data, consistency and dependability of data throughout its lifecycle. In e-commerce and other industries that are dependent on clouds, the integrity of the data is necessary to avoid losing money, reputation and breaking the rules. To address the issue of integrity, organizations have been able to deploy some cloud security mechanisms, including:

- **Encryption:** Cloud data protection is based on encryption, whereby information in a readable form is encoded by using cryptographic algorithms like AES-256 and secure protocols like TLS or mTLS. These guarantee the security of data both during rest and transit. Further separation of encryption keys and data In cloud solutions, envelope encryption (e.g. AWS Key Management Service (KMS) and Azure Key Vault) offers layer-based confidentiality and tampering. This solution also meets regulatory requirements per GDPR, HIPAA and PCI DSS showing verifiable compliance (Wang et al., 2022). As an example, server-side encryption on AWS can be used in an e-commerce system that has KMS to ensure that data on payments is not intercepted or modified.

- **Integrity Checking:** In order to augment the encryption, integrity checks are used to make sure that there has been no change in the data that is stored or being transferred. Hashing (e.g. SHA-256), checksums and digital signatures are common approaches, where each of them generates a distinct fingerprint of data so that any manipulation can be detected immediately. Object versioning and automatic checksum validation are also used in the distributed storage system like Google Cloud Storage to ensure continuous verification and early corruption detection (Haroun & AlKabour, 2019). This verification and encryption combination ensures a combination of both technical soundness and ethical responsibility as it ensures the accuracy and reliability of user-based information.

- **Access Control and Identity Management (IAM):** No less important are the access control mechanisms, which limit the number of people who can access or manipulate sensitive data. **Role-Based (RBAC), Attribute-Based (ABAC) and Policy-Based (PBAC)** models apply permissions, depending on user roles, contextual and network conditions, which reduces the prevalence of insider threats and accidental exposures (Sadavarte et al., 2020). They are enhanced with the **Identity and Access Management (IAM)** solutions that authenticate, authorize, and monitor users using **multi-factor authentication (MFA)**, real-time auditing, and artificial intelligence-based anomaly detection (Olabanji et al., 2024). An analysis of the major cloud vendors shows that AWS is the best at integrating services, Azure has superior Zero Trust and conditional-access capabilities, and GCP has excellent context-aware policies that enhance compliance and accountability.

Feature / Capability	AWS IAM	Azure AD	GCP IAM	Effectiveness / Notes
User & Group Management	Users, groups, and roles; fine-grained policies	Users, groups, roles, dynamic groups	Users, groups, roles, service accounts	Both platforms have hierarchical roles based management; Azure has dynamically grouped automation.
Authentication Methods	Passwords, MFA, AWS SSO, federated login	Passwords, MFA, conditional access, federated login	Passwords, MFA, Identity-Aware Proxy	All support MFA; Azure provides strong conditional access based on risk and location
Authorization / Access Control	RBAC, policy-based, resource-level permissions	RBAC, ABAC, Conditional Access Policies	RBAC, predefined roles, custom roles	GCP is project-level granular, whereas Azure has attribute-based context controls.
Audit & Monitoring	AWS CloudTrail logs all API activity	Azure Monitor & Activity Logs	Cloud Audit Logs	Each of them offers auditing; the AWS and GCP offer wide API-level logs; the Azure can work well with SIEM tools.
Integration with Cloud Services	Deep integration with AWS services	Integrates with Microsoft 365, SaaS apps, and Azure services	Integration with GCP services, BigQuery, Cloud Storage	AWS strongest for its native services; Azure excels in hybrid/multi-cloud; GCP has cloud-

Feature / Capability	AWS IAM	Azure AD	GCP IAM	Effectiveness / Notes
				native automation
Zero Trust / Conditional Access	Limited built-in, requires additional services	Supports Zero Trust policies natively	Supports context-aware access policies	Azure has the most established Zero Trust architecture; GCP has context-aware access; AWS has to be configured.
AI/ML Support for Threat Detection	GuardDuty, IAM Access Analyzer	Microsoft Sentinel AI & conditional alerts	Cloud Security Command Center	They are all AI-assisted threat detection, with anomaly detection and risk scoring being built in with Azure and GCP.

The comparative analysis shows that although both cloud providers have the necessary IAM features, federated identity and adaptive authentication can be used by organizations in a hybrid or multi-cloud environment. The centralized approach of implementing IAM solutions by having one identity broker allows improving policy consistency, administration overhead, and assists in compliance by allowing a central view of all platforms. The emergence of passwordless authentication and decentralized identity (DID) in recent times is transforming IAM by eliminating single points of failure and making credentials user controlled. The combination of these and cloud IAM solutions may increase the levels of trust and interoperability between multi-clouds.

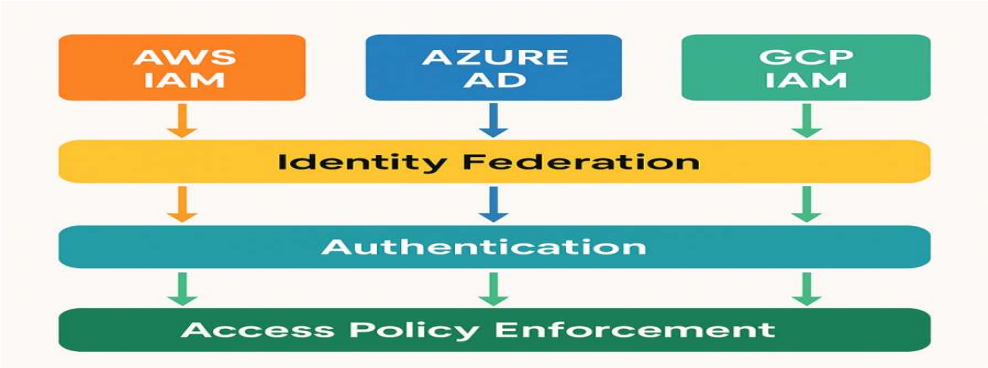

Figure 1: IAM Architecture Comparison

A flow chart of comparing AWS IAM, Azure AD and GCP IAM with identity federation, authentication and access policy enforcement.

- **Critical Evaluation and Synthesis:** As a combination, encryption, integrity checking,

and IAM systems form a multi-tiered protection that implies the secured confidentiality, authenticity, and accountability at every step of the data processing. However, there are issues with their implementation as well. KM distributed, the overhead of the performance due to the frequent encryption and integrity checks, and the effort spent on the organization of IAM positions necessitate a thoughtful strategic planning (Plong-ngooluam et al., 2016; Singh et al., 2023). The companies need to strike a balance between high-level security with usability and system performance using cloud-native services like AWS Security Hub or Azure Defender to manage compliance and monitoring automatically. These mechanisms, together with an ethical and regulatory framework, such as privacy-by-design, staff education, and open governance will not only improve the resilience and regulatory adherence but will also help maintain the customer trust and the existence of long-term data integrity.

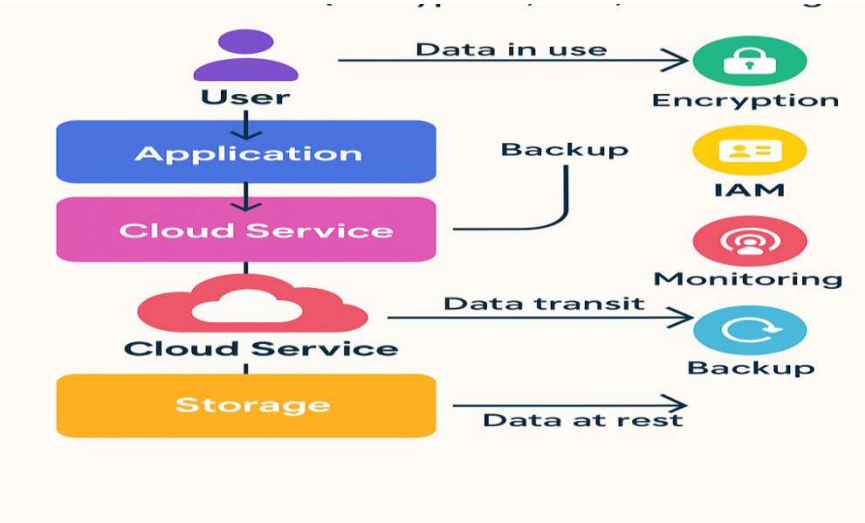

Figure 2: Multi-Layered Cloud Security Architecture (Encryption, IAM, Monitoring)

This number depicts the defence in cloud environment. It demonstrates how data is secured through encryption, identity and access management (IAM), continuous monitoring, and backup methods regardless of the state in which it is used, in transit, and rest.

Task 2: Cloud Security Risks and Threats

2.1 Identification and Discussion of Cloud Security Risks

The issues of the threats to cloud security can be broadly divided into external and internal vectors. Breach, DDoS attacks, and insecure APIs are considered as external threats, and insider misuse or configuration mistakes are the reasons why internal threats may occur. The knowledge of these categories will allow a proactive defensive approach wherein technical protection is coordinated with organizational policies.

Cloud environments have quite a number of risks that are capable of compromising confidentiality, integrity and availability of information. Among the most frequent hazards is the **breach of the data**, and it happens when the unauthorized parties gain access to the information saved or delivered via the cloud-based scenarios, frequently thanks to the weak APIs, ineffective access controls, or improperly configured firewalls (I. Gupta et al., 2022).. In the context of e-commerce where personal and financial information is large and can be abused, these incidences may lead to loss of reputation and legal action. An example of this can be the Capital One breach of 2019, which involved more than 100 million records being exposed after a firewall was misconfigured and the multiple instances of Amazon S3 bucket mis-configuration have revealed how bad governance exposes customer data.

Another challenge that is of utmost concern is the insider **threats**, which occur when loyal employees or contractors deliberately or inadvertently use their access rights in an inappropriate manner. This risk is magnified by the shared-responsibility model since the service providers and the clients are in charge of managing crucial operations. Such vulnerabilities can be addressed by implementing role-based access control and least-privilege policies as well as Zero Trust architectures in which each request is actively authenticated (Hasan, 2024).

No less harmful are **Distributed Denial-of-Service (DDoS)** attacks that flood the network resources and interfere with the provision of the services. The vulnerability of e-commerce businesses is especially high due to the direct impact of the downtime on the sales and trust. Such attacks can be automatically identified and prevented using cloud-native solutions like

AWS Shield and Azure DDoS Protection, and analytical visibility can be provided via the integration of Azure and Sentinel. The best option, though, will be based on the monitoring capacity and budget of an organisation (Soveizi et al., 2023).

The other issue being raised is **insecure APIs**, which will interlink cloud systems, but may be used as entry points by attackers in case authentication or validation is not sound. The application of powerful technologies, including OAuth 2.0, JSON Web Tokens, and ongoing vulnerability testing, will help to make sure that only authorized users and requests can access the system (R. Gupta et al., 2021). Lastly, **information loss** is always a threat due to ransomware, accidental deletion, or lapse in synchronisation. The continuity and compliance of operations as well as SLA are also preserved with the help of robust prevention strategies, including geo-redundant backups, versioning, immutable storage, and regular disasters-recovery tests (Sauber et al., 2021) .

Taken together, these threats underscore the fact that merely having an effective set of technical defences is not enough; it will be essential to demonstrate good governance, transparency, and follow such international standards as GDPR and PCI DSS in order to facilitate the proper risk management. The combination of encryption, IAM, and constant monitoring through a layered, ethical, and proactive strategy is required to ensure resilience and confidence in the cloud ecosystem in the present-day scenario.

Organizations can reduce downtime and business continuity by ensuring that the backup and recovery protocols are resilient.

Threat / Risk	Description	Mitigation / Solution	Example / Tool
Data Breaches	Unauthorized access to sensitive data due to weak controls, APIs, or vulnerabilities	Encryption, strong access controls, regular security audits, regulatory compliance	AES-256, AWS KMS, PCI DSS compliance
Insider Threats	Malicious or accidental actions by trusted users	RBAC, least privilege, Zero Trust Architecture, behavioral analytics	Azure Conditional Access, IAM audit logs
DDoS Attacks	Overloading cloud resources to cause disruption of services	DDoS protection, rate limiting, CDNs, load balancing	AWS Shield, Azure DDoS Protection, Cloudflare

Threat / Risk	Description	Mitigation / Solution	Example / Tool
Insecure APIs	Weak authentication, unverified input, or excessive permissions	OAuth 2.0 / JWT, regular vulnerability testing, API monitoring	API Gateway, Cloud Audit Logs
Data Loss	Accidental deletion, ransomware, hardware failure, disasters	Geo-redundant backups, automated integrity checks, disaster recovery testing	AWS S3 versioning, Azure Backup, GCP Cloud Storage with versioning
Transmission Errors	Data corruption during transfer	TLS / mTLS encryption, checksum verification, integrity checks	TLS 1.3, Google Cloud checksum validation
Unauthorized Modifications	Changes by unauthorized users	IAM policies, audit trails, multi-factor authentication	AWS IAM, Azure AD Conditional Access, GCP IAM

As shown in the summary table above, cloud risk mitigation does not limit to the use of technical tools. The protection should be effective which is achieved through combining these tools with the governance policies, periodical audits, and incident response frameworks. This multi-layered defense is not only minimized to technical vulnerabilities but also formulates the system to fit the shared responsibility model of cloud security.

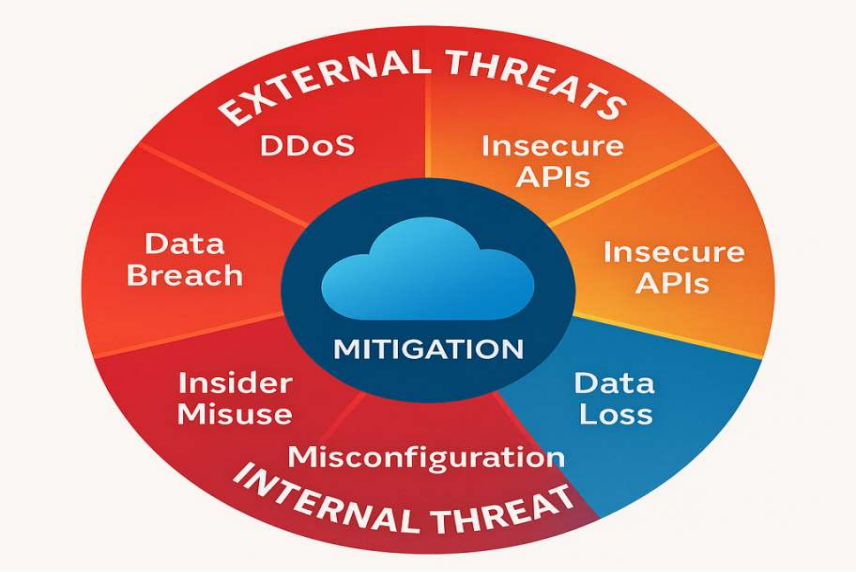

Figure 3: Common Cloud Security Threats and Attack Vectors

I utilize a circular scheme subdivided into the parts: External Threats (DDoS, Data Breach, Insecure APIs) Internal Threats (Insider misuse, Misconfiguration, Data loss).

2.2 Security Solutions for Cloud Environment

Cloud computing has transformed the manner in which organizations store, process and also manage data. Nevertheless, this change also presents great security issues. A complex security solution should be adopted to secure cloud environments. They are encryption, access control, DDoS and network security, data backup and recovery, and live monitoring.

The deployment of a secure cloud environment is based on various complementary mechanisms which ensure the confidentiality, integrity and availability of data throughout the lifecycle of the data. The **encryption** has been the core of the cloud data security since it avoids unauthorized release and interference as well as ensures that the information is secured at rest, transit and use. In data in transit, Transport Layer Security (TLS) and mutual TLS (mTLS) will authenticate both the client and the server against interception or man in the middle attacks. In storing the data, strong algorithms like AES-256 and envelope encryption where the data encryption keys are encrypted with key-encryption keys are used will guarantee resilience even in the case of storage media compromise. In the new field of confidential computing, there are technologies such as Intel SGX and AMD SEV, enabling the processing of data in confidential enclaves without being exposed in plaintext to improve confidentiality in computation. Because deduplicated cloud systems require strong encryption and ownership-verification schemes, according to Dave et al. (2025), these systems are crucial towards preventing unauthorized access, as well as integrity and confidentiality (Dave et al., 2025).

To ensure that unauthorized access does not occur and to eliminate insider threats, **Identity and Access Management (IAM)** is needed to complement encryption. Multi-factor authentication (MFA) introduces the additional authentication of passwords, hardware devices, or biometrics and mitigate the possibility of credential theft and phishing. **Role-based access control (RBAC)** is a system of restriction where access is restricted by job role, and the **attribute-based access control (ABAC)** is one that dynamically removes restrictions on access depending on context (like device, location, and time). **Zero Trust Architecture (ZTA)** model is founded on the basis of the principle of never trust, always verify, which means that every access request is continuously assessed with the help of behavioural and contextual cues. Recent changes also incorporate the use of blockchain to store access logs that cannot be altered to improve

transparency and trust in multi-cloud systems (Punia et al., 2024).

Network and DDoS protection strategies should also be adopted by the organizations to ensure continuity of services. Denial-of-service attacks can lead to loss of revenue and infrastructure failure because they may limit access to cloud servers by overloading them with spam. Volumetric attacks are automatically prevented by cloud-native solutions like AWS Shield and Azure DDoS Protection, which identify abnormalities. Malicious requests such as SQL injections and cross-site scripting (XSS) are blocked using **Web Application Firewalls (WAFs)** and **load balancers and redundant** multi-region deployments are used to provide high availability even under stress. Ometov et al. (2022) mention that the integration of DDoS prevention into the operational process enhances resiliency and reduces the number of manual interventions (Ometov et al., 2022).

The backup and recovery of data are still a crucial part of any cloud-security system. Mechanized backup plans avoid mistakes of human error and integrity-checking tasks ensure that the data is and should be consistent and free of corruption. Backups stored in different geographic locations and version histories enable the organizations to revert to normal operation within a short period after the ransomware or unintentional erasure. The introduction of immutable storage, which is non-editable data after it has been written does not allow making changes after it has been written, and implementing a disaster-recovery test to keep operations going and to comply with the rules and regulations. As noted by Dave et al. (2025), a combination of efficient performance and security in deduplicated storage is possible through efficient ownership-verification that helps in efficient recovery operations (Dave et al., 2025).

In addition to technical implementation, ethical and governance issues are also important. Good security practices can be disclosed transparently, there should be policies on data-ownership, and one should comply with privacy-by-design in the model of shared responsibility. Besides, AI-based tracking systems should be transparent and explainable to ensure the trust of stakeholders. Altogether, encryption, IAM, network defence strategies and data-recovery tactics make a multi-layered and ethically sound solution that improves the resilience of organizations and complies with system frameworks including GDPR and ISO 27001.

In addition to compliance, ethical governance in the cloud should guarantee algorithmic transparency, fairness and accountability particularly as AI and automation become more important in detecting threats and controlling access. To ensure that AI systems do not produce any discrimination, organizations ought to assess them in terms of bias and explainability, and ensure that they comply with cross-border data privacy frameworks, including GDPR and ISO 27001. The combination of privacy-by-design and data sovereignty principles can ensure that cloud resilience is not only developed in a technological aspect, but also moral and socially responsible.

2.3 Real Time Tracking and Analytics.

Although preventive measures are used, it is important to notice the threats at an early stage. Live monitoring and analytics give insight into the activity of the clouds and allows detecting anomalies before they can lead to an incident.

- **Security Information and event management (SIEM):** SIEM systems centralize log data across various sources, correlate logs, and offer significant alerts. Such centralized model allows quicker threats detection and response.
- **Machine Learning (ML): Artificial Intelligence (AI):** AI/ML algorithms are able to identify trends, identify anomalies, and forecast possible attacks. An example is that an AI system might make an alert about an abnormal account of a sign indicating that the account is compromised.
- **Cloud-Native Monitoring Tools:** Cloud APIs and activities like tracking API calls, user activities, and system events can be tracked using such providers as AWS, Azure, and Google Cloud, where they have such tools as CloudTrail, Azure Monitor, and Google Cloud Operations Suite. These tools offer detailed information on the activity of the cloud, which allows compliance audits to be conducted and threats to be detected.

Felemban et al. (2020) show that AI-assisted monitoring architectures are able to minimize false positives and enhance detection accuracy to maximize security operations and minimize operational overhead.

When Security Information and Event Management (SIEM) systems are integrated with automated response systems like Security Orchestration, Automation and Response (SOAR), handling the incident becomes much more efficient. This integration allows real-time isolation of the detected threats, reduces data downtime, and provides data security at all levels data-in-use, data-in-transit, and data-at-rest. Nevertheless, excessive trust in automated analytics may provide an incorrect signal or the detection patterns, so human control is essential to respond responsibly to incidents and form judgment.

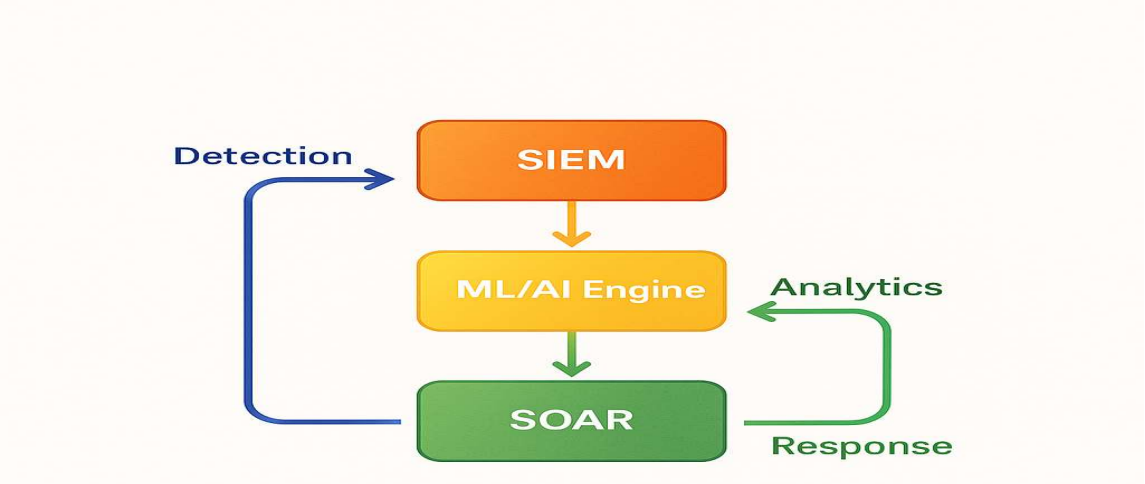

Figure 4: AI-Enableds Real-Time Cloud Monitoring and Incident Response Workflow

The observations provided in the analysis above form the basis of the subsequent recommendations, which enable the strengthening of organizational resilience, ethical responsibility, and technical soundness of cloud operations.

2.4 Recommendations

Organizations implementing a multi-layered and ethically oriented approach to the issue of cloud resilience by integrating Zero Trust verification, regular penetration testing, and employee cybersecurity awareness can help enhance resilience. Active auditing and AI-assisted monitoring facilitate the development of early anomalies, whereas a more profound protection of significant resources is provided by complete encryption and key management through a blockchain. Powerful backup and recovery -with immutable and geo-redundant storage- will ensure that the

storage is available during disruptions. To ensure privacy of information and regulatory conformity in the future, it is advised to implement quantum-safe encryption algorithms like lattice- and hash-based encryption. With the help of the alignment of these efforts to GDPR, ISO 27001 and privacy-by-design, organizations will be able to attain technical excellence and ethical responsibility, enhancing long-term trust of the stakeholders.

Moving forward, quantum-safe encryption algorithms like lattice-based or hash-based encryption algorithms should increasingly be used in future-ready architectures to prepare the disruptive quantum computing capabilities of quantum systems. An initial experimentation in post-quantum key-management will help to assure that cloud infrastructure will be robust even after classical cryptographic life cycles are over.

All these recommendations will create the holistic cloud-security posture. They strengthen the ideals of Zero Trust, constant surveillance and active suppression of threats. With the alignment of such measures with such compliance frameworks as ISO 27001 and GDPR, organizations will be able to ensure the confidentiality, integrity, and availability of cloud resources and enhance customer trust and resilience of business.

Also, the organization must consider a strong governance and ethical framework with privacy-by-design and periodical ethical audits and staff sensitization to monitor accountable cloud security practices and data custodianship.

Conclusion

Cloud computing has revolutionized the way organizations handle and process data, being scalable, flexible and cost effective. Nevertheless, it also poses significant security risks, such as the risks to the integrity of the data, unauthorized access, insider threats, DDoS attacks, insecure APIs, and potential loss of backup of data. It is especially important that the data integrity and security be guaranteed in such industries like e-commerce where sensitive personal and financial

data is processed on a regular basis.

This report has discussed some of the strategies that can be deployed to contain these risks. Encryption, checking the integrity of data, access controls and Identity and Access Management (IAM) systems offer powerful mechanisms to address unauthorized access, data tampering, and corruption. DDoS mitigation, network security, and recovery and backup strategies make systems more resilient and improve the continuity of the business. Also, artificial intelligence/machine learning analytics, real-time monitoring, and security information and event management (SIEM) systems enable organizations to identify unexpected events and act in advance in response to threats.

Effective cloud security requires a multi-layered strategy involving the use of technology solutions, regulatory compliance, employee awareness, and the continuous observation. Although the execution of these solutions poses some challenges to the performance of the computer including overheads and complexity, security, and usability can be balanced with operational efficiency with proper planning and management. Through these steps, organizations will be able to improve their cloud security posture, protect sensitive information, preserve customer confidence, and be compliant with legal and industry regulations.

In general, this report addresses the Learning Outcomes by critically examining the cloud security alternatives to data integrity and identity management and by identifying the risks, layered security solutions, and defining the monitoring requirements of data-in-use, in-transit, and at-rest.

Finally, sustainable cloud resilience is not only possible by deploying encryption or IAM schemes, but is also achieved by Altruism culture of responsiveness, continuous staff training, and active foresight of technological transformation. Through AI-based auditing, privacy-protective analytics and quantum-resistant encryption, organizations can protect the sovereignty of long-term data and still enjoy the goodwill of the population and regulatory peace.

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